

Project Testing Phase

Smart Lender: AI-Powered Loan Approval Prediction System

Testing Approach

The Smart Lender system was tested to ensure accurate loan approval prediction and smooth application performance. Testing included evaluating the Machine Learning model on the test dataset, validating user inputs through the Flask web application, and verifying the complete prediction process from data entry to result display.

Sample Test Cases

Test Case	Input	Expected Result
Valid Applicant	Good credit history, sufficient income	Loan Approved
Invalid Applicant	Poor credit history, low income	Loan Rejected
Missing Field	Applicant Income left blank	Validation Error
Boundary Value	Minimum loan amount	Prediction generated successfully
Valid Input	All mandatory fields entered correctly	Prediction displayed without errors

Model Evaluation Results

Model	Training Accuracy	Testing Accuracy
Decision Tree	1.0000	0.7670
Random Forest	1.0000	0.7849
K-Nearest Neighbors (KNN)	0.8425	0.7061
Gradient Boosting Classifier	0.9204	0.7921

Testing Outcome

- All input fields were validated successfully.
- The Flask web application accepted valid applicant details and generated predictions correctly.
- Invalid or incomplete inputs were handled without crashing the application.
- The **Gradient Boosting Classifier** achieved the highest testing accuracy and was selected as the final prediction model.
- The complete loan prediction workflow functioned successfully from user input to result display.